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Partition panel system

Abstract

A partition system configured for building small temporary structures. The system includes a series of panels, each having cooperating connection interfaces. The connection interfaces include one or more moveable hooks on a first panel that is configured to extend into and attach a neighboring panel. The system also includes right angle frame members that allow for panels to be connected at right angles.

Inventors:	Fisher; Nathan (Monsey, NY)
Applicant:	Fisher; Nathan (Monsey, NY)
Family ID:	92898483
Assignee:	Fisher; Nathan (Monsey, NY)
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Primary Examiner: Mintz; Rodney

Attorney, Agent or Firm: Tarter Krinsky & Drogin LLP

Background/Summary

FIELD OF THE INVENTION

(1) The invention relates to the field of partition panels, more particularly to a panel system that can form partition wall or small rooms and structures.

BACKGROUND OF THE INVENTION

(2) There are many different needs for temporary booths, rooms, room partitions, both indoor and outdoor. Known partitions are not sufficiently sturdy, nor easy to assemble. There is therefore a need in the art for partitions formed of lightweight, yet sturdy materials and which can be easily assembled and disassembled.

SUMMARY OF THE INVENTION

(3) Embodiments of the invention provide a partition system formed of panels having a key-locking interfaces. The panels of the invention are lightweight yet insulated allowing for the rapid assembly and disassembly of the same.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1 is a front perspective view of a series of panels in accordance with an embodiment of the invention.
- (2) FIG. 2 is a broken-away partially exploded view of a panel and frame section in accordance with an embodiment of the invention.
- (3) FIG. 3 is a top partial cross-sectional view of a panel and frame member in accordance with an embodiment of the invention.
- (4) FIG. 4 is a top partial cross-sectional view of a panel and frame member in accordance with an embodiment of the invention.
- (5) FIG. 5 is a top partial cross-sectional view of two connected panels showing interlocked frame members in accordance with an embodiment of the invention.
- (6) FIG. 6A is a schematic view of a key positioned to lock a panel whereby a hook is shown in a stored position in accordance with an embodiment of the invention.
- (7) FIG. 6B is a schematic view of a key positioned in a keyhole and rotated whereby the hook is shown in a locked position.
- (8) FIG. 6C is a partial view of a left side of a panel showing a bar configured to receive the hook shown in FIG. 6B in accordance with an embodiment of the invention.
- (9) FIG. 7 is a partial front view of connected panels schematically showing a hook deployed in a locked position in accordance with an embodiment of the invention.
- (10) FIG. 8 is a top cross-sectional view of a corner frame member connected to a panel in accordance with an embodiment of the invention.
- (11) FIG. 9 is a perspective view of a panel having a corner frame positioned for connection.
- (12) FIG. 10 partial perspective view of panels connected at a right schematically showing a hook deployed in a locked position in accordance with an embodiment of the invention.
- (13) FIG. 11 is a perspective view of a walled structure formed in accordance with embodiments of the invention, with cross-beams and respective hangers show in exploded view in accordance with embodiments of the invention.
- (14) FIG. 12 is a perspective view of a walled structure formed in accordance with embodiments of the invention.
- (15) FIG. 13 is an enlarged view of a hanger joined to a panel shown supporting a crossbeam in accordance with an embodiment of the invention.

DETAILED DESCRIPTION OF THE INVENTION

- (16) Embodiments of the present invention will now be described with reference to the above-identified figures of the drawings. However, the drawings and the description herein of the invention are not intended to limit the scope of the invention. It will be understood that various modifications of the present description of the invention are possible without departing from the spirit of the invention. Also, features described herein may be omitted, additional features may be included, and/or features described herein may be combined in a manner different from the specific combinations recited herein, all without departing from the spirit of the invention.
- (17) With reference to FIG. 1, a series of attachable panels are shown. Each panel is similarly constructed. For example, with reference to panel 14, an outer layer face plate 16 is shown surrounded by a frame 18 formed of right, left, upper and lower frame members. In the orientation shown, right frame member 18a is provided with keyholes 20 on its inward facing side, whereas a left frame member 18b is provided with pockets on its left side wall. Pockets 22 are configured to receive a hook emanating from a neighboring panel. As shown, panel 24 is similarly provided with pockets 22 on its left side wall, whereas, a right frame member 18a is provided with keyholes 20.
- (18) FIG. 2 shows an enlarged broken-away exploded view of a panel and frame section. As

shown, panel (e.g. **14**) is shown formed of a first outer layer **16** (e.g. face plate), and a parallel second outer layer **18** (e.g. back plate). Face plate **16** has a first outward facing side **26** and an inward facing side **28** bridged by plate thickness **27**. Back plate **18** similarly has an outward facing side **30** and inward facing side **32** bridged by thickness **31**. In embodiments of the invention, a core material **34** is disposed between face plate **16** and back plate **18**. In embodiments of the invention, core **34** is an insulative material such as corrugated, foam, fiberglass, cellulose or the like. In embodiments of the invention, inward facing side **28** of face plate **16** and inward facing side **32** of back plate **18** are adhered to the core material, for example, via glue or similar adhesive.

(19) With continued reference to FIG. 2, a side aspect of face/back plates **16**, **18** and core material are shown. As shown, side aspect **36** is formed of side thicknesses **27**, **31** of face/back plates **16**, **18** and core material **34** sandwiched therebetween. It will be understood that right side aspect is similarly constructed.

(20) In embodiments of the invention, outward facing side **26** of face plate **16** and outward facing side **30** of back plate **18** are provided with channels. In embodiments of the invention, channels are provided on respective face/back plates **16**, **18** offset from the outer perimeters thereof. For example, FIG. 2 shows a channel **38** offset from the left edge of panel **14**. Channel **38** is an indentation on face/back plates **16**, **18** that extend into thickness **27**, **31**. As shown, in embodiments, a mirroring channel **40** is made in back plate **18**. In an embodiment of the invention, channels **38**, **40** extend the entire length of face/back plates **16**, **18** or a majority thereof.

(21) In embodiments of the invention, as mentioned, each panel is provided with a surrounding frame. With continued reference to FIG. 2, a broken away view of a left frame member **18b** is shown. As shown, frame member **18b** is formed of a first wing **42** and a second substantially parallel wing **44** bridged by spine **46**. First wing **42** has an outer surface **48** and an inner surface **50**. Second wing **44** similarly has an outside surface **52** and inside surface **54**. Spine **46** has an outside surface **56** and an inside surface **58** and a thickness **60**. In embodiments of the invention, a rib **62** is provided on inside surface **50** of first wing, and a mirroring rib **64** is provided on inside surface **54** of second wing. In embodiments of the invention, ribs **62**, **64** are integral extensions of inner surfaces **50**, **54** of first and second wings **42**, **44**. In embodiments of the invention, ribs **62**, **64** are sized and shaped for insertion into respective channels **38**, **40**.

(22) As best shown in FIG. 3 channel (e.g. **64**) is formed of a first angled wall **68** extending from inner surface **54** of wing **44** and a second angled wall **70**. In embodiments of the invention, the second angled wall **70** is disposed at close to a 90° angle to inner surface **54** of wing **44**. In embodiments of the invention second angled wall **70** is disposed between an 80° and 90° angle from inner surface **54**. In this regard, second angled wall **70** is either parallel to, or substantially parallel to first side wall **72** of channel **40**. A substantially mirroring construction is provided with respect to rib **62** and channel **68** of face plate **16**.

(23) In embodiments of the invention, and as shown in FIG. 4, a decorative or protective sheet may applied to cover face plate **16**, back plate **18** or portions thereof. For example, as shown, a sheet **71** is applied to outward facing side **30** of back plate **18**. A similar sheet **73** is applied to inward facing side **26** of face plate **16**. Sheets **71**, **73** may formed of any fabric or similar material such as leather, vinyl, plastic or such similar materials. In embodiments of the invention, sheets, **71**, **73** are adhered to back plate **18** or front plate **16** via glue or similar adhesive. In other embodiments of the invention, sheets **71**, **73** are not adhered via adhesive, rather, inside surface of respective wings (e.g. **44** and other wings of respective frame members) maintain a sheet (e.g. **71**) in place.

(24) In construction of panels, frame members (e.g. **18b**) are forced onto panels such that ribs **62**, **64** become seated in respective channels **38**, **40**. For example, as shown in FIG. 3, once frame member **18b** is applied, inner surface **50** of first wing contacts outward facing side **26** of face plate **16**, inner surface **54** of second wing **44** contact outward facing side **30** back plate **18**, and inside surface **58** of spine **46** contacts side aspect **36** of face/back plates **16**, **18** and core material **34**.

(25) In embodiments of the invention, inside surface **58** of spine **46** is provided with a series of

indentations **74** that run the length of inside surface **48**. Indentations **74** provide additional surface area for glue or adhesive to bind inside surface **58** of spine to side aspect **36** of face/back plates **16**, **18** and core material **34**. In embodiments of the invention, glue is only used to bind inside surface **58** of spine to side aspect **36** of face/back plates **16**, **18** and core material **34** and no glue is used between respective wings **42**, **44** and face/back plates **16**, **18**. This allows for a tight connection of the frame—but also allows wings **42**, **44** to move slightly away from face/back plates **16**, **18** when pried by a user, for example, to add or remove a decorative sheet as described.

(26) With continued reference to FIG. **3**, in embodiments of the invention outside surface **56** of spine **46** is formed of tongue and groove elements. For example, as shown, tongue **76** is a substantially square or rectangular-shaped protrusion formed of an extension of thickness **60**. Tongue **76** is shown substantially parallel to a second tongue **78**. Space between tongues **76**, **78** forms groove **80** that is substantially the same size and shape of tongue **76**. Any number of tongues and grooves are possible in different embodiments of the invention.

(27) In embodiments of the invention, frame members are positioned on left and right sides of panels in opposing orientations. In this regard, tongues on a first panel will align with grooves on a second panel so that they may be joined in an interlocking manner. For example, FIG. **5** shows right frame member **18a** of a first panel connected to a left frame member **18b** on a second panel. As shown, respective tongues (e.g. **76**, **78**) intermesh with respective grooves (e.g. **80**).

(28) In embodiments of the invention, in addition to the intermeshing of tongues and grooves as described, panels are secured to each other by one or more retractable hooks. For example, with reference to FIG. **6A** a panel (e.g. **14**) is shown having a right pocket **82** or similar groove in its right side wall. In embodiments of the invention, right pocket **82** is a carved-out section of core material that is surrounded with a metal frame **84**. Right pocket **82** houses a pivotable hook **86** that is connected to an axle or similar pivot. In embodiments of the invention, the axle upon which hook **86** is affixed is rotatable (clockwise and counterclockwise) by a key **88** or similar tool and is accessible via keyhole **20**.

(29) As shown in FIG. **6C**, left side wall of panel (e.g. **24**) is provided with a left pocket **90**. In embodiments of the invention, left pocket **90** is a carved-out section of core material that is surrounded with a metal frame **92**. Left pocket **90** houses a bar **94** that is positioned between a first pocket wall **92a** and second parallel pocket wall **92b** (i.e. positioned orthogonally to hook **86**). Bar **94** is a substantially cylindrical member that is sized and shaped to receive hook **86**. As shown in FIG. **6B** key **88** is utilized to move hook **86** from a first stowed position to a second deployed position (i.e. engaging bar **94**).

(30) FIG. **7** shows a partial front view of two panels (e.g. **14**, **24**) where hook **86** is shown in a first stowed position A, and in a second locked position B. In the locked position, hook **86** surrounds a portion of bar **94** thereby locking two panels (e.g. **14**, **24**) together.

(31) It will be understood that each panel is provided with both a right pocket housing a rotatable hook (e.g. **82**) and a left pocket (e.g. **90**) housing a bar or similar catch. It will be further understood that pockets could be reversed (e.g. a pocket housing a hook on left and a pocket housing a bar on left). It will be understood that each panel is provided with at least one right pocket and one left pocket (housing a hook and bar, respectively), however, other numbers of pockets are possible, for example, each side of a panel may be provided with two, three, four etc. pockets.

(32) It will be understood that pockets provided on respective panels are positioned at substantially identical height positions to pockets on neighboring panels. For example, with reference to FIG. **1**, panels **14**, **24** are shown, each having three pockets on its left side and three pockets on its right side. Each respective pocket is provided on the same level as a neighboring pocket on a neighboring panel. For example, a middle pocket (e.g. **82**) on panel **14** is positioned substantially at the same height level as neighboring pocket **90**. The pockets above and below the middle pockets **82**, **90** are similarly configured. In this regard, when a first panel (e.g. **14**) is placed side to side

with a second panel (e.g. **24**) all respective pockets align, providing a user one more points of connection. In the embodiment shown in FIG. **1**, three points of connection are provided.

(33) In assembling panels in accordance with embodiments of the invention, a user aligns two panels such that respective pockets align. The user then secures the panels together by rotating each of the respective hooks from the stowed position to the locked position.

(34) In embodiments of the invention, specialized panels are provided having frame members that are configured for connecting a neighboring panel at a right angle. This allows for corner panels to be connected.

(35) For example, referring to FIG. **8**, a top cross-sectional view of a corner frame member **96** is shown. Corner frame member **96** is formed of a first wing **98**, a substantially parallel second wing **100**, and a bridging spine **102**. Wing **98**, **100** are configured like wings described herein (e.g. **42**, **44**), for example, each having inward facing ribs to grip respective channels as described. An extension wall **104** extends away from wing **98**. Substantially parallel to extension wall **104** is a panel-engaging frame section **106** which substantially in line with wing **100**. Extension wall **96** and panel engaging frame section **106** are bridged by a substantially orthogonal end wall **108**. As shown, panel-engaging frame section **106** is positioned, substantially at a right angle to the side aspect of panel **107**.

(36) In embodiments of the invention, panel-engaging frame section **106** is formed of tongues and grooves as described herein and also is provided with pockets containing bars, also as described herein. For example, as shown in FIG. **9**, panel **110** left frame member (in the orientation shown) is a corner frame member. Panel engaging frame section **106** is shown having three pockets (e.g. **112**), each having a bar **94** configured to receive corresponding hooks emanating from respective pockets on panel **114**. This allows a connection of one panel (e.g. **110**) at a substantial right angle to a second panel (e.g. **114**).

(37) FIG. **9** shows a perspective view of panels **110**, **114** joined at a right angle. Hook **86** is shown locked around bar **94** of pocket **112**.

(38) Panels of the invention may be used to form walls of any of various configurations, including straight walls, L-shaped walls, and U-shaped wall and four-walled structures. For example, FIG. **11** shows a four-walled structure formed by joined panels in accordance with embodiments of the invention.

(39) It will be understood that panels in accordance with embodiments of the invention may be provided with windows **116**, doors **118** or the like. In this regard, the outer perimeters of a panel (e.g. **120**, **122**) having a window/door include the connection features described herein (tongue and groove interface and respective pockets), but a central portion thereof is provided with a window/door.

(40) Embodiments of the invention also provide a roof beam system that is used to install cross beams over an upper opening of a structure. As shown in FIG. **11**, hangers **124** are provided having an upper hook **126** and a lower hook **128**. Upper hooks **124** are sized and shaped to grip an upper edge of panels, whereas, lower hooks **128** are configured to connect to a cross beam **130**.

(41) FIG. **12** shows a structure with respective cross beams **130** hung by opposing hangers **124** provided on opposite walls. It will be understood that any of various coverings may be applied atop respective cross beams in different embodiments of the invention.

(42) While this invention has been described in conjunction with the embodiments outlined above, it is evident that many alternatives, modifications and variations will be apparent to those skilled in the art. Accordingly, the exemplary embodiments of the invention, as set forth above, are intended to be illustrative, not limiting. Various changes may be made without departing from the spirit and scope of the invention.

Claims

1. A panel system, comprising: a first panel comprising a face plate, a back plate and core material disposed between the face plate and the back plate; the first panel having a front facing surface, a rear facing surface and a right side wall; the first panel comprising a top frame member, a bottom frame member, a left frame member and a right frame member surrounding the first panel; a pocket disposed on the right side wall comprising a rotatable hook; a second panel comprising a face plate, a back plate and core material disposed between the face plate and the back plate; the second panel having a front facing surface, a rear facing surface and a left side wall; the second panel comprising a top frame member, a bottom frame member, a right frame member and a left frame member surrounding the second panel; and a pocket disposed on the left side wall comprising a bar configured to receive the rotatable hook, wherein each face plate comprises a first channel extending at least a portion of a length thereof, and each back plate comprises a second channel extending at least a portion of a length thereof, wherein each of the right and left frame members comprise a first wing having an inside surface and an outside surface, a second wing having an inside surface and an outside surface, and a spine having an inside surface and outside surface, the spine interconnecting the first wing and the second wing, and wherein the inside surface of the first wing comprises a first rib that is configured to be inserted in the first channel and the inside surface of the second wing comprises a second rib that is configured to be inserted into the second channel.
 2. The panel system of claim 1, wherein the outside surface of the spine comprises tongues and grooves.
 3. The panel system of claim 1, wherein the pocket disposed on the left side wall comprises a first pocket wall and a second pocket wall, wherein the bar is secured between the first pocket wall and the second pocket wall.
 4. The panel system of claim 1, wherein the rotatable hook disposed in the pocket of the first panel is movable into the pocket disposed on the second panel.
 5. The panel system of claim 1, wherein the rotatable hook is disposed on an axle.
 6. The panel system of claim 5, wherein each right frame member comprises a keyhole.
 7. The panel system of claim 6, wherein the axle is accessible through the keyhole.
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